Isaac Alexander Vazquez

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Education

California Polytechnic State University *San Luis Obispo, CA*
Bachelor of Science in Computer Engineering *Expected Dec 2026*

- **Relevant Courses:** Digital and Computer Design, Microcontrollers and Embedded Applications, Real-time Operating Systems, High-Performance Embedded Systems, Autonomous Mobile Robotics, Hardware Security

Skills

- **Programming languages:** C, C++, C#, Python, Java, SystemVerilog, RISC-V Assembly
- **Hardware Platforms:** FreeRTOS, Docker, Linux, ROS2, Oscilloscope, Multimeter, GazeboSim, Foxglove
- **Microcontrollers:** STM32 ARM-Cortex Microcontrollers, NVIDIA Jetson Nano Dashboard, ESP32
- **Languages:** Spanish (fluent)

Technical Projects

Amiga Robot Driver Configurations | *ROS2, NVIDIA Jetson, CAN Bus* *Aug 2026*

- Joined an existing four-wheeled agriculture robotics project and re-evaluated drivers built by a previous team to assess performance.
- Sniffed CAN packets between a PC and a Jetson Nano, analyzing frames with ROS2 Linux commands and oscilloscope measurements of voltage changes.
- Documented debugging processes for future collaborators and created launch files to start multiple ROS2 nodes concurrently.

Gobilda Mobile Robot | *ROS2, LiDAR, Foxglove, NVIDIA Jetson, GobildaSim* *Aug 2026 - present*

- Developed ROS 2 Python nodes on an NVIDIA Jetson to command linear and angular wheel velocities via a serial-interfaced ESP32 microcontroller using UART.
- Implemented Finite State Machine (FSM) and behavior tree logic for real-time obstacle avoidance, utilizing LiDAR sensor data parsed at a deterministic 10 Hz loop rate.
- Validated control algorithms and sensor pipelines using Foxglove Studio telemetry alongside GoBildaSim simulation environments prior to physical hardware deployment.

RTOS Traffic Light System | *FreeRTOS, Embedded Systems, STM32* *June 2026*

- Re-architected a bare-metal traffic light system into a multi-task FreeRTOS application handling button input and asynchronous interrupts.
- Replaced polling and blocking delays with interrupt-driven button handling, using FreeRTOS APIs for precise timing and inter-task resource management.
- Debugged over UART using PuTTY and used software timers with callbacks to reduce task overhead.

Function Generator | *Embedded Systems, Timer Interrupts, SPI, STM32* *Oct 2025*

- Designed and debugged an STM32L476RG system that generates AC waveforms with user-selectable shape, duty cycle, and frequency.
- Implemented SPI communication (master mode) to drive an external 12-bit DAC for waveform output.
- Achieved 100-500 Hz output from keypad input using polling and timer interrupts, verified on an oscilloscope.

Work Experience

ITS Student Assistant *San Luis Obispo, CA*
Cal Poly Information Technology Services *May 2025 – Present*

- Provide technical support to students, faculty, and staff across hardware, software, and network issues, resolving about 10 issues per week.
- Image and deploy 5+ Windows and macOS devices for faculty and staff, installing applications and configuring campus network connectivity IPV4.
- Document troubleshooting steps from client tickets in Jira, creating a reference used by student assistants and new hires.

Involvements

Society of Hispanic Professional Engineers (SHPE) *San Luis Obispo, CA*
Member *Aug 2022 – Present*